

REBALANCING-EQ: The Variant Matrix – Findings and Analysis

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1 Purpose and lineage

The **rebalancing-eq** scenario family joins the two simulation programs that had previously run apart: the **equilibrium campaign** (`alberta_buck/sim/EQUILIBRIUM.md` – the BUCK-K monetary loop on real 5-year macro data, judged by the parity-with-interior-K gate) and the **rebalancing machinery** (the BasketRebalanceDirector and the BuckBasket as a savings vehicle, previously exercised only on synthetic 3-token regimes). Every arm below runs the eq-baseline financial structure – the 6-token M2-laggard weighted basket (LABR 2400 / CNST 2200 / FOOD 2000 / NRG 2000 / PAXG 800 / cbBTC 600 bp), the real macro window, prorata basket, K0 0.75 with rails (0, 0.95), seed 0xA1BC – through real contract deployments on an EVM (pyrevm), with the director keeper active, at full rebalancing cadence (day_step 1, ticks 4: daily whale pins, 6-hour PID samples).

Acceptance gate (unchanged from the campaign): tail (last 25%) basket-ValueInBuck within 1.00 +/- 0.02, K never railed in the tail, issuance/redemption channels alive.

2 The family: variants, running, reproducing

Experiments: `alberta_buck/sim/experiments/rebalancing-eq-5yr*.toml`. Variant extension names compose deltas left-to-right on three axes – (1) knobs (`[agents.*]` and `[deploy]`), (2) trend regime (reserved), (3) disruption (`[[interventions]]` price shocks). Each TOML is standalone (no cross-file inheritance) and its resolved config is embedded in the output vector's meta.

- `make nix-sim-rebalancing-eq-<ext>` runs one arm -> `test/vectors/rebalancing-sim-eq-<ext> + images/rebalancing-sim-eq-<ext>.png`;
- `make nix-venv-sim-rebalancing-eq-matrix` re-runs EVERY arm against the **current contracts** (recompiles first, MATRIX_JOBS-way parallel) and prints the gate table – the "adjust the BuckBasket, observe the net effects" loop;
- `make nix-sim-compare-rebalancing-eq` gates existing vectors only.

The final vectors are ~25-30MB each and are NOT committed; the rendered images, this document, and the one-command matrix regeneration are the record. Runs are deterministic: same code + same TOML -> identical vector.

3 Model rulings (decisions of record)

These calibration and semantics decisions were made deliberately and should not be silently re-litigated:

1. **Per-day agent rates.** Rate knobs (saver `base_rate`, borrower re-tire/release rates, the funding factor's issue-rate signal) are per-DAY, scaled by `day_step/ticks_per_day` each activation, so daily flow is cadence-invariant. The coarse macro mode (`day_step`
 - (a) is an approximation for slow dynamics only; the notorious cold-start buckUsd spike of early runs was substantially an artifact of 10-day PID sampling and lumpy per-act flows.
2. **Atomic refinance** (`refi_mode="atomic"`, this family only; the audited octl configs keep "paced"). A debtor evaluates the WHOLE mortgage conversion ex ante: quote the BUCK needed to net the full USDC mortgage after slippage and pool fee, check issuance capacity (spendable headroom + K-scaled unactivated face), gate the theta law on the all-in execution discount, then execute BuckCredit $\rightarrow$ BUCK $\rightarrow$ USDC $\rightarrow$ full payoff in one act – or do nothing (no mint, no premium paid, re-quote next month). The paced model committed first and discovered the market later; it also made every deployed debtor a perpetual net BUCK absorber (monthly save-rate buffer accumulation with one-shot issuance), which is what strangled the early debtor arms.
3. **Admission frontier $LTV \leq K$ is expected.** A mortgage above `face * K` cannot be fully funded by BUCK issuance; in reality the borrower would top up with supplemental USDC credit. Those borrowers are IGNORED for now (declined debtors are fully inert). `K` is thereby the emergent admission gate on mortgage conversion – wide when the system wants supply, narrow when it does not.
4. **Housing-market distribution.** The home (the insured asset) is the primary draw – `home_k` [200,1500] LOG-uniform (median $\sim$ \$530k), mirroring real estate's right skew – and the rest derives: `face` = the appraised home value (the insurable value the BuckCredit attests; premium charged on it; `creditLimit` = `face*K`), `mortgage` = `home * ltv` with `ltv` [0.05,0.95] (mean $\sim$ 0.49, the seasoned-ladder average), `income` = `home * [0.20,0.35]`. The legacy independent mortgage/face draws implied a maximally-leveraged civilization (LTV massed near

1.0 with underwater tails) and excluded $\sim 88\%$ of the population at the frontier.

5. **Deep exit route.** The floating BUCK/USDC pool is \$50M family-wide ([deploy] `target_buck_lp_m = 50`; `deploy.py` now sizes the pool off its documented knob). A national-scale currency pair is deep, and a shallow pool was an artificial exit bottleneck; real best-cost routing would additionally spread exits across BUCK/TOKEN $\rightarrow$ TOKEN/USDC legs. The BUCK-unaware arbs keep the deep pool balanced. With atomic refinance, this depth is the literal admission control on credit adoption ($\sim 2\%$ all-in on a $\sim \$1\text{M}$ mortgage vs $\sim 10\%$ at $\$10\text{M}$).
6. **Saver endowment** (`arrive_mode="endow"`): pre-window BUCK inventory transferred from SimLP against a zero-premium credit. NB: minting on the saver's own credit endows K-scaled **headroom** (balanceOf counts unused creditLimit, which melts as K falls), not held BUCK; the third-party transfer is what makes it real inventory (totalSupply == sum of positive signed balances).
7. **Endogenous arrivals** (`arrive_mode="endog"`): debtor hazard = $(K/k_{\text{ref}}) * (1 + \text{arr_gain} * (1 - \text{bvib}))$ capped at `arr_max`; depositor hazard = $\text{dev_ref} / \max(\text{dev_ref}, \text{ewma}(|\text{bvib} - 1|))$. hazard == 1 reproduces the open-loop "originate" stagger exactly, so the endog-vs-originate comparison isolates price-responsiveness itself.

4 The final matrix (2026-08-19/20)

All ten arms PASS. Zero railed frames in any tail; zero execution errors; every debtor in every arm eventually completed a full atomic refinance (declines were transient re-quotes – the frontier acted as a queue, not a wall).

arm	bv tail (sd)	K	iss/ret \$M	refis	notes
(banked eq-baseline, coarse cadence, \$10M ub)	0.9926 (0.0265)	0.819	1.73/1.52	–	the re
base	1.0005 (0.0107)	0.760	1.60/0.00	–	fine c
capacity	1.0017 (0.0266)	0.634	2.05/1.18	–	lowes
endow	1.0005 (0.0101)	0.715	1.40/0.54	–	inven
deep	1.0006 (0.0101)	0.757	1.38/0.00	–	~ = b
endow-deep	1.0003 (0.0097)	0.715	1.49/0.55	–	== e
debtors	1.0002 (0.0081)	0.763	1.06/0.56	24/24	staggy
debtors-mature	0.9999 (0.0080)	0.925	0.00/1.56	24/24	conve
basketeers	0.9998 (0.0244)	0.860	2.60/1.15	24/24	K-im
originate	1.0006 (0.0072)	0.942	0.51/0.02	30/30	both
endog	1.0002 (0.0068)	0.924	0.43/0.00	30/30	same

5 What each variant does (effect analysis)

5.1 base – the fine-cadence result

The control loop closes with no help: parity to 0.05% with K settled at 0.760. K climbs during the FatCredit adoption S-curve and plateaus exactly when cap_frac saturates – the climb is the controller compensating for **scheduled** capacity withholding, not chronic deflation. Residual weakness: the retirement channel is silent (0.00) – nothing in this cast ever wants to deleverage.

5.2 capacity (savers2x demand + borrower face/funding headroom)

Buys the LOWEST resting K (0.634): with demand and capacity headroom the plant answers the controller easily and the controller retains the most unused authority. The cost is the widest parity spread of the non-basketeer arms (0.0266) – strong saver demand is itself a perturbation. Choose this shape when rail margin is the objective.

5.3 endow / deep / endow-deep – steady-state initialization, attributed

At the original \$10M floating pool, depth did nearly all the work. With the family now at \$50M, the attribution INVERTS: deep (extra TOKEN/BUCK depth) is indistinguishable from base (K 0.757 vs 0.760), while the saver endowment carries the whole remaining effect – endow and endow-deep are

identical (K 0.715, retirement 0.54/0.55 vs 0.00). Standing private sell-side inventory lowers the resting K by ~ 0.045 and is the difference between a one-sided and a breathing channel. Lesson: liquidity depth and distributed inventory are complements, and once one is adequate the other is the binding one.

5.4 debtors (staggered arrival over the first year)

24 honest mortgage refinancers tighten parity (sd 0.0081, from 0.0107) and open the retirement channel, at base-like K. The conversion flow itself is stabilizing: each atomic refinance is a supply event that lands when the market can bear it (theta law + $LTV \leq K$ + quoted slippage), and the ex-debtors' subsequent surplus investment adds basket demand.

5.5 debtors-mature (everyone converts at t_0)

The "converted civilization": all 24 refinance immediately, and the economy then runs on the refinanced supply. K rests HIGH (0.925) – the controller holds the admission gate wide – but interior and settled ($dK \sim 0$), with the second-tightest parity (0.0080) and a tail that is pure voluntary deleveraging (iss 0.00 / ret 1.56M). Under the OLD rulings (paced, shallow, narrow-face) this exact arm railed 81% of its tail; the reversal is entirely the three structural fixes.

5.6 basketeers (+ DirectMint depositor population)

The basket finally exercised as a savings vehicle inside the loop: ~ 950 depositor round-trips, the first treasury accrual, the K-immune 100%-LTV tranche at $\sim 10\%$ of supply. Cost: the loosest debtor-arm spread (0.0244) – depositor churn is real noise the director must absorb – but the gate still passes comfortably. The stylized fear that the K-immune tranche strangles the controller does NOT materialize at this scale.

5.7 originate (+ continuous debtor and depositor origination)

Both sides of the economy grow at roughly matched dollar rates; all 30 debtors convert; the K-immune tranche reaches 13.7M ($\sim 18\%$ of supply) with 1,158 round-trips – and parity is the second-tightest in the program (0.0072). The caution: K rests at 0.942, EIGHT BASIS POINTS of lever

below the 0.95 rail. A growing economy consumes controller authority; see recommendations.

5.8 endog (originate with price-responsive arrival hazards)

The headline theoretical result: the endogenous economy finds essentially the SAME equilibrium as the hand-tuned calendar (bv 1.0002 vs 1.0006, sd 0.0068 vs 0.0072) with slightly MORE rail margin (K 0.924 vs 0.942), and nobody chose a growth rate. Debtor arrivals accelerate into lean markets, depositors wait out demonstrated instability, and the on-chain brakes (funding-factor toll, pre-buy reserve, K-scaled limits) price what arrivals may do. Matched supply/demand growth is an emergent property of the price system, as designed.

6 Cross-cutting findings

1. **Execution semantics and market microstructure matter as much as the controller.** Nothing about the PID changed between the all-FAIL debtor matrix and the all-PASS one – what changed was HOW agents execute (atomic vs paced), WHAT the exit route can bear (\$50M vs \$10M), and WHO the population is (housing distribution vs maximal leverage).
2. **K’s resting level is the admission-frontier position.** A converted or growing economy settles K high because the controller is holding the conversion gate open; that is a feature – but it spends rail margin, which is the robustness budget.
3. **The K-immune basket tranche is tolerable at ~18% of supply.** The fence’s motivating problem did not bite at realistic scale; whether it bites at 30-50% remains the open fence-in-the-loop question.
4. **Private inventory is the retirement channel.** Every arm whose savers hold standing BUCK shows two-sided channels; every arm with empty savers shows one-sided issuance.

7 Recommendations: robustness against economic disruption

Ranked by expected information per unit of compute.

1. **Multi-seed sweep first.** Everything above is seed 0xA1BC. The equilibrium campaign's own history (3/5 seeds failed its plain baseline) is the warning: run the matrix's key arms (base, endow, debtors-mature, endog) across the campaign's 5 seeds via sweep.py before treating any conclusion as robust.
2. **Buy back rail margin for the growth arms.** originate/endog rest 0.008-0.026 below KMAX; any demand shock rails them immediately. Three levers, in order of principle: (a) implement the **top-up borrower** (BUCK for face*K, supplemental USDC for the remainder) so the admission frontier is not the only supply valve; (b) widen the collateral base (new insured asset classes = new face arriving independent of housing); (c) a KMAX policy study – 0.95 was chosen to keep a railed K solvent and observable, but a converted economy may warrant a higher ceiling with an explicit solvency argument.
3. **Exercise the disruption axis.** The naming scheme reserves it; the machinery exists ([[interventions]] price_shock, uptake_shock, re-tune). Priority shocks against the winning arms: (a) a commodity spike (NRGC +30%, cbBTC +50% – the campaign's shock-price recipe) mid-run; (b) a **demand run**: mass depositor exit (exit_ptick shock on the DM population) – the basket-run scenario; (c) a rate-cycle shock (apr shift changes every theta threshold and the refinance arbitrage itself).
4. **Test the oversupply flood.** The feared scenario – massive issuance through unencumbered collateral driving K to its LOWER rail – is still untested. Add carry issuers (BuckIssuerArbAgent exists) or a debtor draw_mult beyond encumbrance matching, and measure: how deep K dives, where bv's discount equilibrates (the theta/ff boundary predicts $\sim \theta \cdot \text{apr}$), and the recovery half-life (K throttles flow, not stock; only Jubilee melt and buy-backs shrink the overhang).
5. **Fence-in-the-loop.** Swap --basket fence into the basketeer arms so basket issuance is K-scaled rather than K-immune, and compare stability AND depositor economics against these baselines. This is the designed join of the fence program with the monetary loop, and becomes urgent if the tranche share is pushed past $\sim 30\%$.
6. **Slow-trend regimes.** grow/decline commodity drifts (the trend axis) need a price-generation hook or staged PriceOverlay ramps – the reverting-

vs-trending distinction mattered enormously in the synthetic program and is unexplored here.

7. **Population churn.** Port the campaign’s population-churn experiment (lost borrower + saver) to this family; the endog hazards should recruit replacements – verifying that is exactly what they are for.
8. **Exit-route realism.** The single deep pool abstracts best-cost multi-route exits (BUCK/TOKEN -> TOKEN/USDC). Teaching the debtor quote (and the arbs) to split across routes would let depth be stressed without the abstraction absorbing it.

8 Artifacts

- Vectors: `test/vectors/rebalancing-sim-eq*.json` (uncommitted; regenerate with the matrix target). Superseded-run checkpoints (paced, shallow-ub, narrow-face eras): `build/sim/runs/partial*`.
- Images: `images/rebalancing-sim-eq*.png` (committed) – the 11-panel render per arm.
- Progress tooling: a watcher script (session scratchpad) renders `build/sim/progress/` every 10 minutes during long runs; the sims checkpoint their vectors every 25 sim-days, so partial runs are always plottable.
- Parent documents: `alberta_buck/sim/EQUILIBRIUM.md` (the campaign), `BASKET-REBALANCE.md` / `BASKET-REDESIGN.md` (the synthetic basket program).